

Stability of U_5Si_4 phase in U-Si system: Crystal structure prediction and phonon properties using first-principles calculations

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ABSTRACT

U-Si systems have recently received considerable attention due to the potential application of U_3Si_2 as a high-density fuel under an accident tolerant fuel initiative. However, the thermodynamic stability of the more recently reported adjacent U_5Si_4 phase is uncertain and could play a significant role in fuel performance. In this work, the enthalpy of formation of the phase predicted by density functional theory (DFT) using the DFT + U formalism is used with an evolutionary algorithm (USPEX) to evaluate stability and possible atomic structures for U_5Si_4 . The structure of U-Si convex hull phases and the confirmed U_3Si_2 structure were predicted providing confidence in the reliability of the evolutionary algorithm, as well as to obtain a convex hull with comparable enthalpies of formation. Subsequently, the code was applied to the U_5Si_4 composition for cells with 18 and 36 atoms, predicting a 36-atom hexagonal symmetrized unit cell with space group $P6/mmm$ as the lowest-energy configuration, agreeing with that experimentally reported for U_5Si_4 . Yet, phonon calculations using the density functional perturbation theory formalism, demonstrated that the predicted structure is dynamically unstable, exhibiting negative vibrational modes for the uranium. These indicated that generated shears are directed toward the formation of potential uranium octahedral sites, analogous to those occupied by carbon atoms in $U_{20}Si_{16}C_3$. It was thus concluded that omitting the U_5Si_4 phase from assessed U-Si phase equilibria is currently justified.

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1. Introduction

Following the Fukushima-Daiichi nuclear accident, an international effort began in the nuclear community to devise a safer fuel-cladding system, which should have a better response in severe accident conditions. Such accident tolerant fuels (ATF) should perform better than the Zirconium alloys- UO_2 system in terms of reduced oxidation kinetics, reduced heat of oxidation in steam, and lower release of hydrogen [1]. A key factor requirement is the replacement of the Zirconium alloy cladding. Some considered alternative cladding candidates are silicon carbide ceramic matrix composite (SiC-CMC), FeCrAl-based ferritic alloys, and Mo-based alloys. However, metallic candidates come with a significant neutronic penalty, precluding the use of UO_2 fuel near current enrichment levels [2], and thus encouraging development of higher uranium-atom density fuel phases. Although SiC-CMC possesses

better neutronic properties than even zirconium (0.086 b compared to 0.20 b) [3], issues such as modulus, swelling, thermal expansivity force the adoption of a large gap between fuel and cladding, to avoid negative mechanical interactions. Thus, even for SiC-CMC cladding use it may be important adopt a fuel with better thermal conductivity. Thus, attractive ATF fuel forms should possess higher uranium atom-density and thermal conductivity, yet also exhibit a high melting temperature, and acceptable resistance to oxidation. A current candidate being explored is U_3Si_2 , long used as a fuel in research reactors operating at relatively low temperature with the phase dispersed in an aluminum matrix. For the significantly more demanding application as light water reactor (LWR) ATF, there remains the need to further understand the behavior of a monolithic pellets under irradiation and accident/transient conditions [4].

The growing application in research reactors in the last decades, has motivated more thorough studies of U-Si phase space [5–8]. The current phase diagram indicates seven intermetallic compounds; USi_3 , USi_2 , $USi_{1.88}$, U_3Si_5 , USi , U_3Si_2 and U_3Si . A U_5Si_4 phase was only reported in 1998 by Noël et al. in a conference

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communicate [9] where a reinvestigation of U–Si binary system in the region U_3Si_2 – USi was performed. Based on x-ray diffraction, Noël et al. [9] proposed that U_5Si_4 phase exhibits a hexagonal unit cell ($P6/mmm$ space group) with lattice parameters $a = 10.468 \text{ \AA}$ and $c = 3.912 \text{ \AA}$ (Fig. 1). Although the work of Noël et al. presents a fully crystallographic description, the reported data is limited and the x-ray diffraction pattern has never been reproduced.

More than a decade later, Berche et al. [10] also reports a U_5Si_4 phase based only on SEM-EDS analyses. Considering the observed microstructure, a formation mechanism through a peritectic reaction [$\text{U}_3\text{Si}_2 + \text{liquid} \rightarrow \text{U}_5\text{Si}_4$] was proposed. However, Berche et al. could not accurately determine the temperature of this peritectic transition, and no peak was observed in differential thermal analysis (DTA), therefore this phase was not included in their thermodynamic assessment. The inclusion of U_5Si_4 in a phase diagram was considered only in the most recent thermodynamic assessment performed by Wang et al. [11]. However, due to limited available information, no crystal structure model was provided, and the Neumann-Kopp rule was needed to describe its properties. It was emphasized by Wang et al. [11] that more work is still required to confirm the stability of the U_5Si_4 phase. Ultimately, understanding U–Si phase equilibria is important as during burnup U:Si ratios can vary promoting formation of additional silicide phases, and as U_5Si_4 would be compositionally adjacent to the U_3Si_2 fuel phase it could indeed be generated as a minor phase influencing fuel performance.

Recently, Middleburgh et al. [12] performed DFT + U calculations for the U_5Si_4 phase using the structure reported by Noël et al. [9]. The formation energy was calculated considering the reaction $\text{U}_5\text{Si}_4 \rightarrow 2\text{USi} + \text{U}_3\text{Si}_2$, with a -0.02 eV reaction energy. Middleburgh et al. [12] note that this small formation energy could be negated by entropic factors, thus no conclusion could be made about U_5Si_4 phase stability.

In this work we use the enthalpy predicted by DFT + U [13], as the fitness function for the evolutionary algorithm Universal Structure Prediction Program (USPEX) [14] to perform a structure search for the U_5Si_4 composition. Comparison of energies of formation for the same material system but with various crystallographic variants allows a broader consideration of the phase stability.

In the present work, USPEX [14] was applied in three steps: (i) Variable-composition calculation for full range in U–Si phase space (a maximum number of 20 atoms per unit cell, and up to 2500 structural optimizations were performed), (ii) fixed-composition calculations for the known U_3Si_2 phase (10-atom unit cell), and (iii) fixed-composition calculation for U_5Si_4 with 18- and 36-atom unit cells. The first sets of calculations were for assessing the

reliability of the evolutionary algorithm, as well as to obtain a convex hull with comparable energies of formation for comparison with respect to the thermodynamic stability of U_5Si_4 . Phonon calculations were also performed to evaluate the dynamic stability of the U_5Si_4 phase, which are particularly important given the likelihood of a small formation reaction enthalpy as noted above.

2. Methods

We used the Vienna ab initio Simulation Package (VASP) [15,16] code to perform the DFT calculations, using the projector-augmented wave (PAW) method. The electronic exchange and correlation energies were calculated within the generalized gradient approximation (GGA) in the Perdew–Burke–Ernzerhof (PBE) formalism [17]. The valence electrons explicitly treated in the calculations were $6s^2 6p^6 6d^2 5f^2 7s^2$ for U; $3s^2 3p^2$ for Si with a plane wave expansion cut-off energy of 600 eV for. Following previous efforts to optimally handle the strong correlation of the uranium 5f electrons [18], we used the DFT + U formalism with the Dudarev [19] implementation of the Hubbard correlation where $U_{\text{eff}} = 1.50 \text{ eV}$. A $\mathbf{k}$ -point spacing ($2\pi \times 0.06 \text{ \AA}^{-1}$) was used to generate Monkhorst–Pack $\mathbf{k}$ -point grids for Brillouin zone sampling [20]. Calculations proceeded self-consistently until the total energy converged to within 0.1 meV/cell, conjugate gradients enthalpy minimization proceeded until the enthalpy changes became smaller than 1 meV/cell. No symmetry was imposed, allowing the volume (ISIF = 3), ionic positions and shape of the cells to vary during the relaxation.

The DFT + U energies were used as the fitness function of the evolutionary algorithm USPEX [14]. The first-generation structures were created randomly, and a population of $N+10$, with N being the number of atoms in the unit cell, was employed in this work. The best 60% of each generation were used to produce the next-generation structures by heredity.

Global optimizations were performed for at least 25 generations after the minimum energy is found. In any mode, no initial assumption about the crystal structure of the known phases were used. All phases were calculated with the same U_{eff} value, which allowed for direct comparison of the calculated enthalpies of formation. The final structural parameters reported were obtained from the FINDSYM [21] algorithm, using a tolerance of 0.05 Å.

Phonon calculations were performed using density functional perturbation theory (DFPT) [22] as implemented in the PHONOPY code [23] for the lowest energy structure of the U_5Si_4 phase, as well as for the neighboring U_3Si_2 and USi phases.

3. Results and discussion

3.1. U–Si convex hull

Evolutionary variable-composition simulations were performed over the full composition range of the U–Si phase space. The optimization of the 2500 structures resulted in the convex hull shown in Fig. 2, where the enthalpies of formation are calculated relative to Si-diamond and α -U. The indicated stable structures/phases are U_3Si , USi , USi_2 and USi_3 with details provided in Table 1.

The predicted structures of the U_3Si , USi_2 and USi_3 phases are in good agreement with the experimental information [24], considering the β - U_3Si phase. The computed symmetry of the USi phase, although not the same as that experimentally observed, is very similar, i.e. #63 rather than #62. While the observed USi –Pmna structure was instead calculated to lie $\sim 0.03 \text{ eV}$ above the U–Si convex hull, it represents a small difference with respect to the computed Cmc structure, and could be the result of the assumed DFT + U yielding a metastable states [25], or even due to entropic

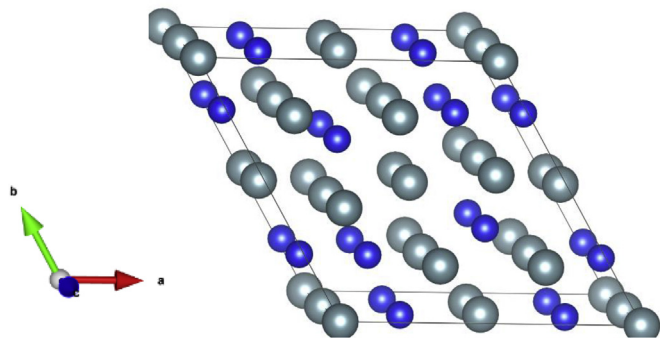


Fig. 1. U_5Si_4 crystal structure proposed by Noël et al. [9] based on x-ray diffraction analysis (uranium and silicon atoms are shown in grey and blue, respectively). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

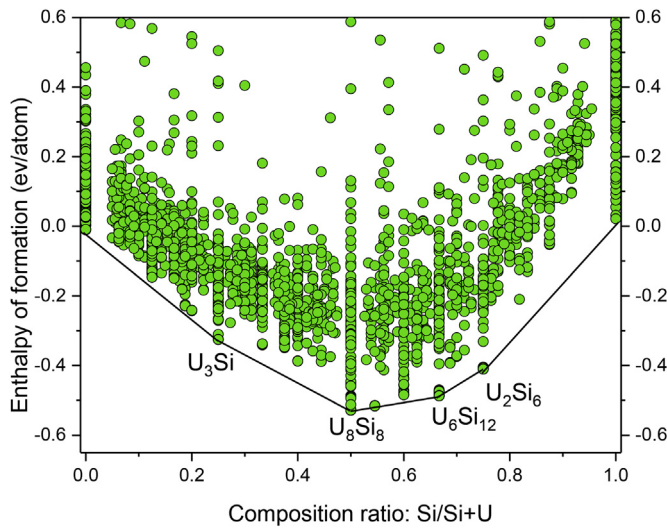


Fig. 2. Convex hull for the U-Si phase space obtained from variable-composition USPEX calculations made in (enthalpies relative to Si-diamond and α -U, with $U_{\text{eff}} = 1.5\text{eV}$), for 2500 structure optimizations.

Table 1

Structural properties of the stable U-Si compounds found from the convex hull for evolutionary calculation up to 2500 structures.

Compound	Space group	a (Å)	b (Å)	c (Å)	Exp. Sym [24].
U ₃ Si	#221, Pm-3m	8.61	8.60	8.60	α - I4/mcm β - Pm-3m
USi	#63, Cmc	7.90	3.84	5.74	Pnma
USi ₂	#191, P6/mmm	4.06	4.06	3.87	P6/mmm
USi ₃	#221, Pm-3m	4.05	4.05	4.05	Pm-3m

contributions, consideration of which were outside the scope of this study. It should be noted that the overall shape of the obtained convex hull is in good agreement with that one recently reported by Wang et al. [11]. The lowest energies found for USi and USi₂ are in agreement with experimental and other DFT results [18].

Two experimentally observed phases [24] are not on the calculated convex hull, U₃Si₅ at 62.5at. Si and U₃Si₂ at 40at. Si. The U₃Si₅ phase is known to have the same structure type as USi₂ (space group *P6/mmm*) [24], the structures varying due to partially occupied Si sites. As partial occupancies cannot be captured by the USPEX code it is absent from the computed U-Si phases. That is not the case for U₃Si₂ which has a well-defined lattice. The composition statistics (Fig. 3), however, indicate the number of structures generated for the 40at. Si composition is low (<50 candidates), which indicates that USPEX code may require more extensive statistics to capture it as U₃Si₂ possesses a more complex crystal structure compared to the other phases. Thus, it may need more generations to capture the structural complexity. While other low statistics stoichiometries are occurring, the focus is only on previously reported phases in the U-Si system. A large number of generations, as well as an increased unit cell size (>20 atoms) is likely necessary to explore new possible phases. Regardless, the results reported here demonstrate satisfactory performance of USPEX in computing phase stability in the U-Si system using DFT + *U*.

3.2. Fixed-composition U₃Si₂

As with any evolutionary approach, as noted above it is expected that the results will improve with more a larger numbers of generations and thus to increase the statistics for the 40at. Si

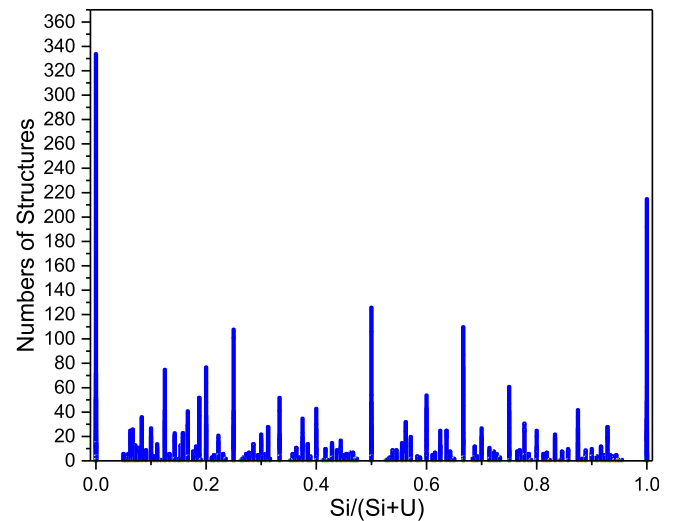


Fig. 3. Composition statistics for the variable-composition USPEX calculations U-Si for 2500 structures.

composition, fixed-composition calculations were performed. As the U₃Si₂ structure is well-known experimentally, this step was to evaluate the efficiency and reliability of the USPEX code for predicting the structure of USi_x phases at fixed compositions. The size of cell adopted here has 10 atoms and 25 generations halting criterion was imposed, which means the calculation will stop only if the same minimum energy structure persists for at least 25 generations. The energies and structural properties of the are displayed in Fig. 4 and Table 2, respectively.

The USPEX evolutionary algorithm closely predicted the experimental structure as the ground state for the 40% at. Si composition, with its enthalpy of formation being on the U-Si convex hull, reflecting the stability of this phase. Comparing the experimental structure to the predicted one [26], the latter slightly overestimates the parameters by ~1.5% for *a* and *b* and ~3.3% for *c*. These values represent satisfactory agreement considering the use of the GGA + *U* approximation.

It is important to note that a minimum of 400 variants were necessary to reach the correct atomic structure, confirming the hypothesis that the larger and more complex the unit cell, the more generations necessary to reach the ground state. This result demonstrates the capability of the evolutionary algorithm to find a global minimum energy structure, and emphasizes the importance of using adequate numbers of generations to be successful.

3.3. Fixed-composition U₅Si₄

For the U₅Si₄ composition, structural searches were performed with two distinct sizes of unit cells, 18 and 36 atoms. The calculations extended to 1500 and 3000 structures, respectively. The computed enthalpies of formation together with those obtained from computing the U-Si convex hull and the structure proposed by Noel et al. [9] are shown in Fig. 5.

For the calculation with 36 atoms unit cell, the evolutionary algorithm found the most stable configuration a hexagonal structure with space group *P6/mmm* with lattice parameters of *a* = 8.06 Å and *b* = *c* = 10.60 Å. This structure is similar to the one reported by Noel et al. (*a* = 10.468 Å, *c* = 7.82 Å) [8], with similar overestimation of lattice parameters as for U₃Si₂ phase. Both structures are energetically located very close to the calculated U-Si convex hull, indicating that this phase may become thermodynamically stable above 0 K. The obtained density for the U₅Si₄ phase

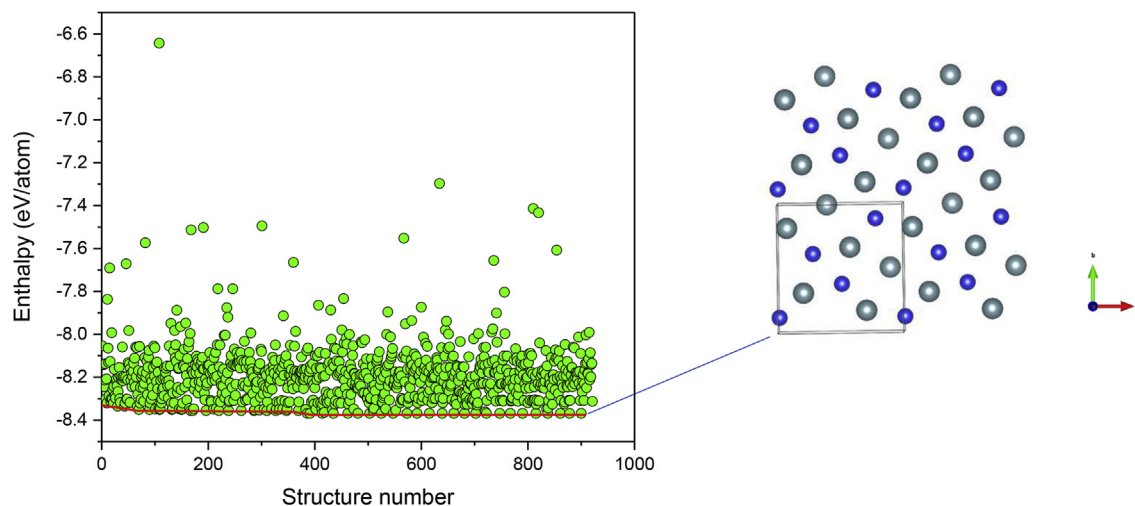


Fig. 4. Enthalpy of U_3Si_2 structures (per 10 atoms), the best individuals per generation are along the red line. The most stable U_3Si_2 structure found by the USPEX calculations is shown in left side. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 2

Structural properties of the most stable U_3Si_2 structure found from USPEX calculations.

Compound	Space group	a (Å)	b (Å)	c (Å)
U_3Si_2 (USPEX)	#127, $P4/mbm$	7.44	7.44	4.03
U_3Si_2 exp [24].	#127, $P4/mbm$	7.3299	7.3299	3.9004

is 11.01 gm/cm^3 in comparison with 11.49 gm/cm^3 for the U_3Si_2 , which implies that if this phase forms locally in the fuel matrix, it should not cause a significant volume change.

The calculation with 18 atoms per unit cell returned a different structure, with a more positive enthalpy of formation, and therefore is not considered in the further analysis of the U_5Si_4 phase.

3.4. Dynamical properties of U_5Si_4 structure

Phonon frequencies need to be assessed to find if increasing temperature beyond 0 K can change the relative position of the U_5Si_4 phase with respect to the U-Si convex hull and to determine the dynamical stability of the U_5Si_4 phase.

The calculated phonons dispersions for U_5Si_4 ($P6/mmm$), U_3Si_2 ($P4/mbm$) and USi ($Pmna$) phases are shown in Fig. 6. The results demonstrate the presence of imaginary phonon frequencies at the Γ -point for the U_5Si_4 phase, indicating this phase is dynamically unstable. In contrast, the U_3Si_2 and USi phases, calculated for the same conditions, show no imaginary modes.

To better understand the source of the dynamic instability for U_5Si_4 , we examined the phonon modes with negative frequency (Fig. 7). It is evident that the instability for U_5Si_4 structure is related to the uranium site atom. The displacements necessary to stabilize the structure are toward uranium octahedral sites, with contraction in the c-direction and a small expansion within the a-b plane (shown by the red arrows in Fig. 7). A similar, but thermodynamically stable ternary phase is found in the U-Si-C system, $U_{20}Si_{16}C_3$ [27]. This ternary phase has the same structure type as the U_5Si_4 (hexagonal $P6/mmm$), where the carbon atoms occupy the uranium octahedral sites, the same sites that are formed by the shears necessary to eliminate the negative frequencies for U_5Si_4 . This implies that the U_5Si_4 structure may be stabilized by substitutional light element dopants like oxygen or carbon impurities on the uranium octahedral sites, and that the experimentally reported phase could be the result of such.

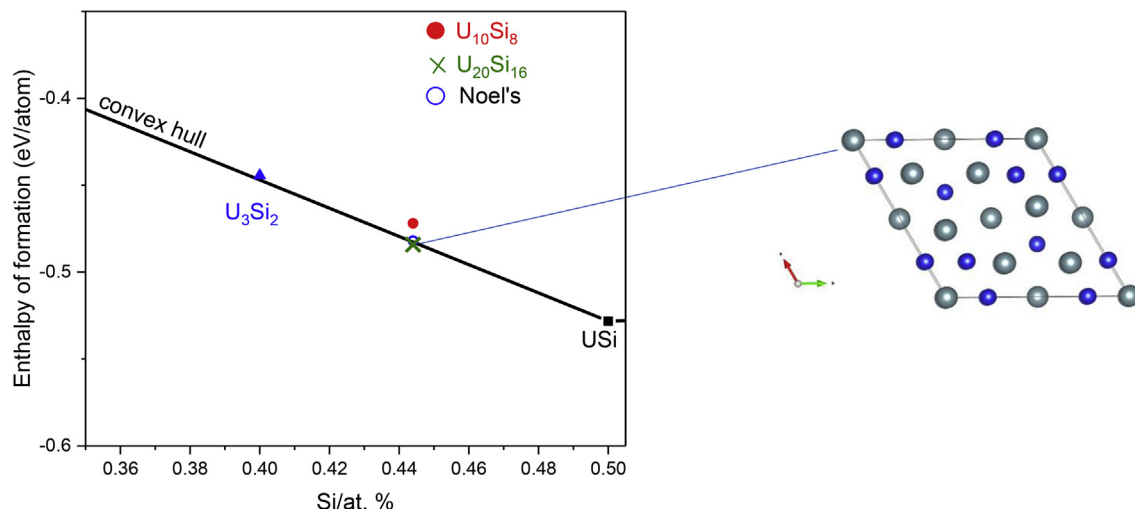


Fig. 5. Enthalpy of formation for the predicted lowest energy structures of U_5Si_4 using $U_{10}Si_8$ and $U_{20}Si_{16}$, and the experimentally reported structure of Noel et al. [9].

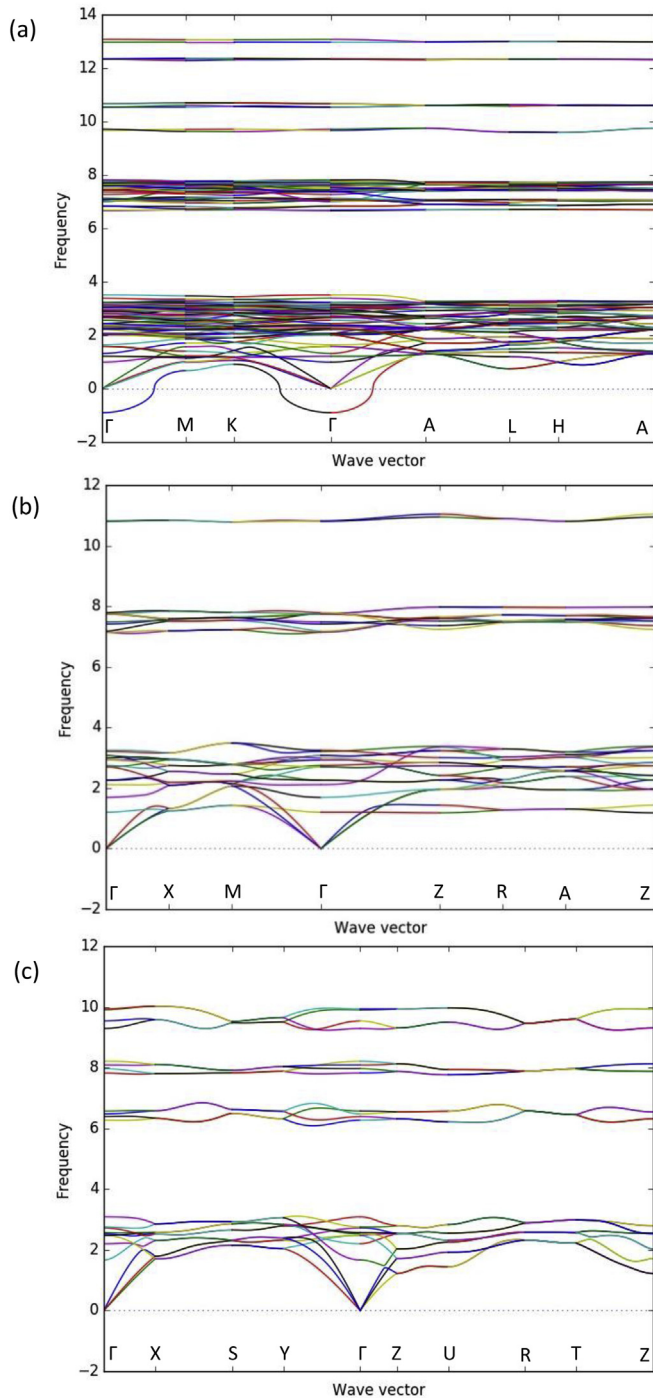


Fig. 6. Phonons dispersion curves showing dynamical instability of the U_5Si_4 $P6/mmm$ (a) phase and dynamical stability of U_3Si_2 (b) and USi (c).

4. Conclusions

The stability of a reported U_5Si_4 phase was evaluated using first principles calculations. The evolutionary algorithm USPEX was used to obtain the U-Si convex hull, results compared against known phase equilibria, and employed to explore the energy landscape of the U_5Si_4 composition. The results demonstrate good reliability for USPEX code in phase predictions in the U-Si system, although large statical ensembles are required to capture complex phase properties, i. e. larger unit cells with larger numbers of

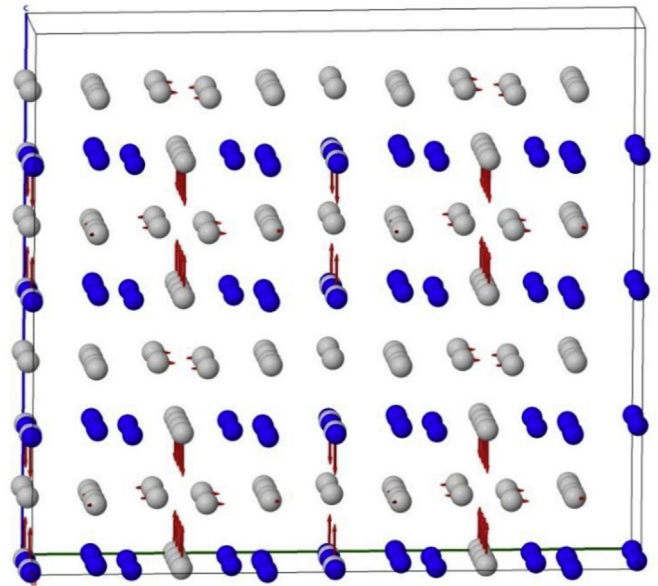


Fig. 7. Atomic displacement directions, shown by red arrows, associated with the most negative eigenvector of the dynamical matrix at the Γ -point in the $2 \times 2 \times 2$ supercell. Uranium and silicon atoms are shown in grey and blue, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

symmetrically unique atoms. A hexagonal $P6/mmm$ structure, with an enthalpy of formation very slightly above the calculated convex hull was found as the lowest energy configuration at the U_5Si_4 composition. However, phonon calculations in the DFPT formalism reveal that this structure is dynamically unstable. The phonon modes with negative frequency occur at the uranium atom sites, implying that these stabilize the structure. This together with the observation the existence of a stable isostructural ternary, $U_{20}Si_{16}C_3$ ($P6/mmm$) phase implies that the experimentally reported U_5Si_4 could be stabilized by contaminant oxygen or carbon. Given the nature of the computational results, and confidence in them gained from assessing known phases, together with the implication of dopant stabilization of the phase, and the inconsistent experimental observation of U_5Si_4 , there is ample justification for not including the U_5Si_4 phase in the assessed U-Si system.

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